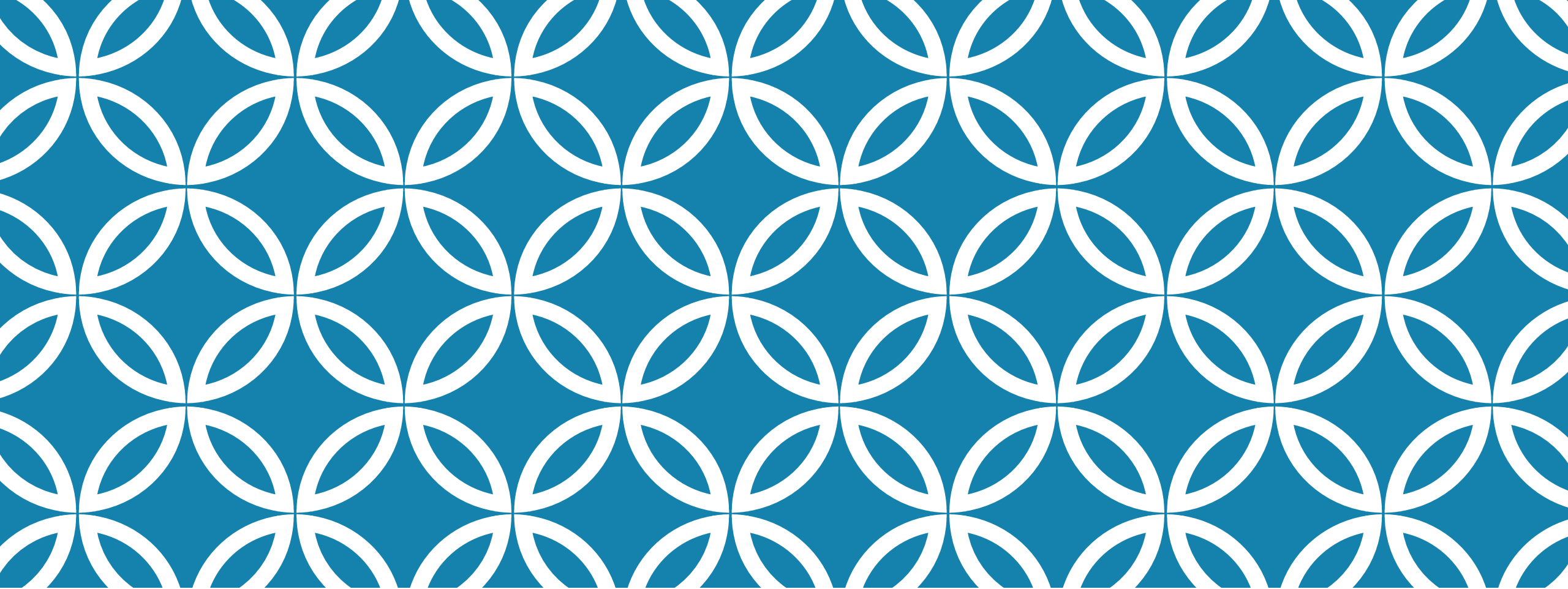




CSG553 RECONFIGURABLE COMPUTING

Networked Embedded Systems & Robotic Applications Group
(NETRA)

Dr. Meetha V Shenoy
Associate Professor,
BITS Pilani, Pilani Campus
Lect-1

COMPUTING UNIT

The requirements vary according to the Target applications

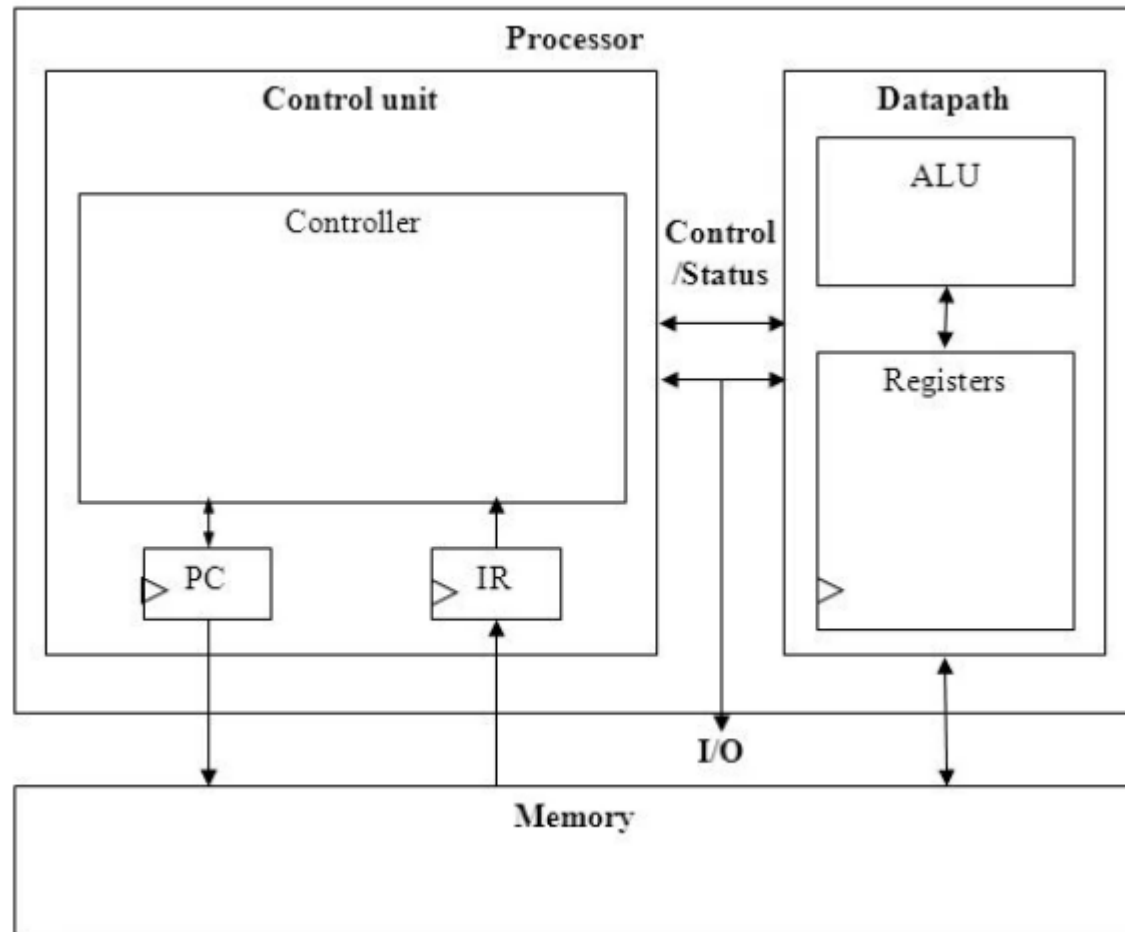
- Performance
- Power
- Price of the final equipment
-

Programmability of the system

VON NEUMANN COMPUTER

In 1945, the mathematician Von Neumann (VN) demonstrated in study of computation that a computer could have a simple structure, capable of executing any kind of program, given a properly programmed control unit, without the need of hardware modification

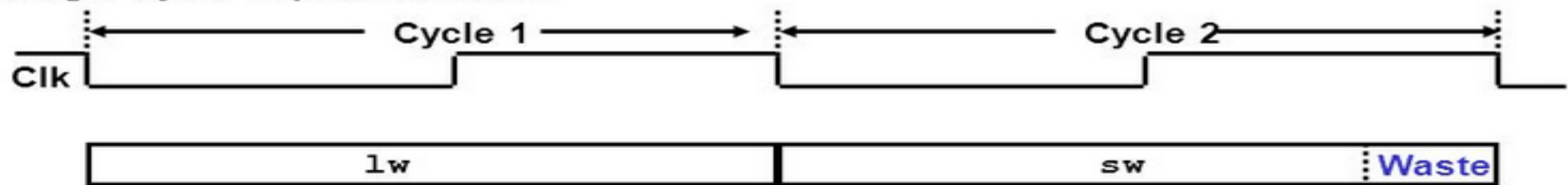
CPU INTERNAL ARCHITECTURE



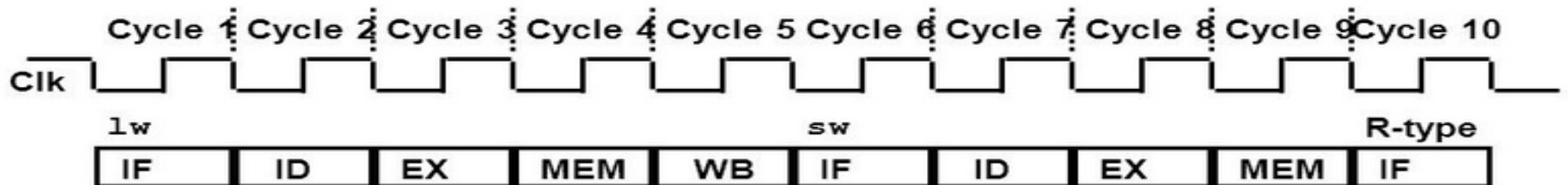
Stored program and data model

SINGLE CYCLE VS MULTICYCLE VS PIPELINED

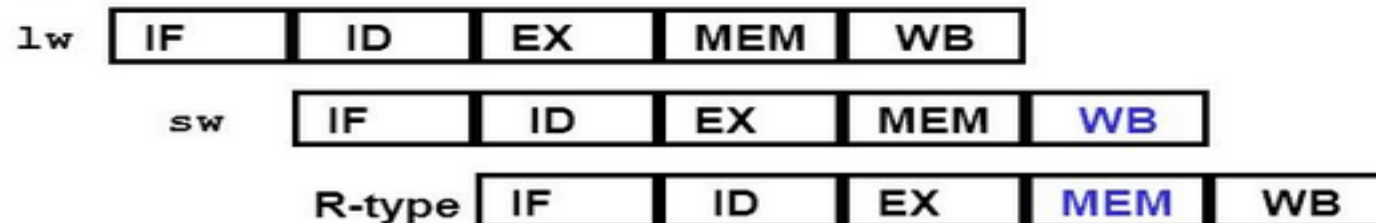
Single Cycle Implementation:



Multiple Cycle Implementation:

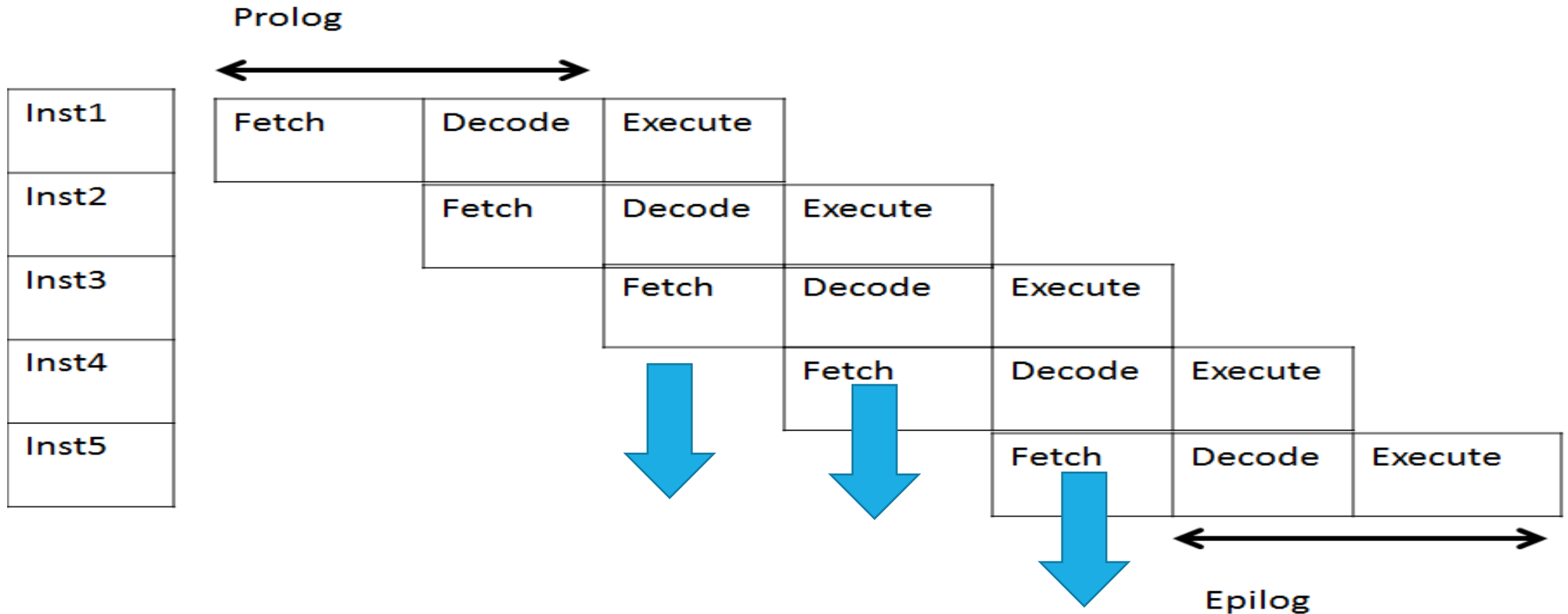


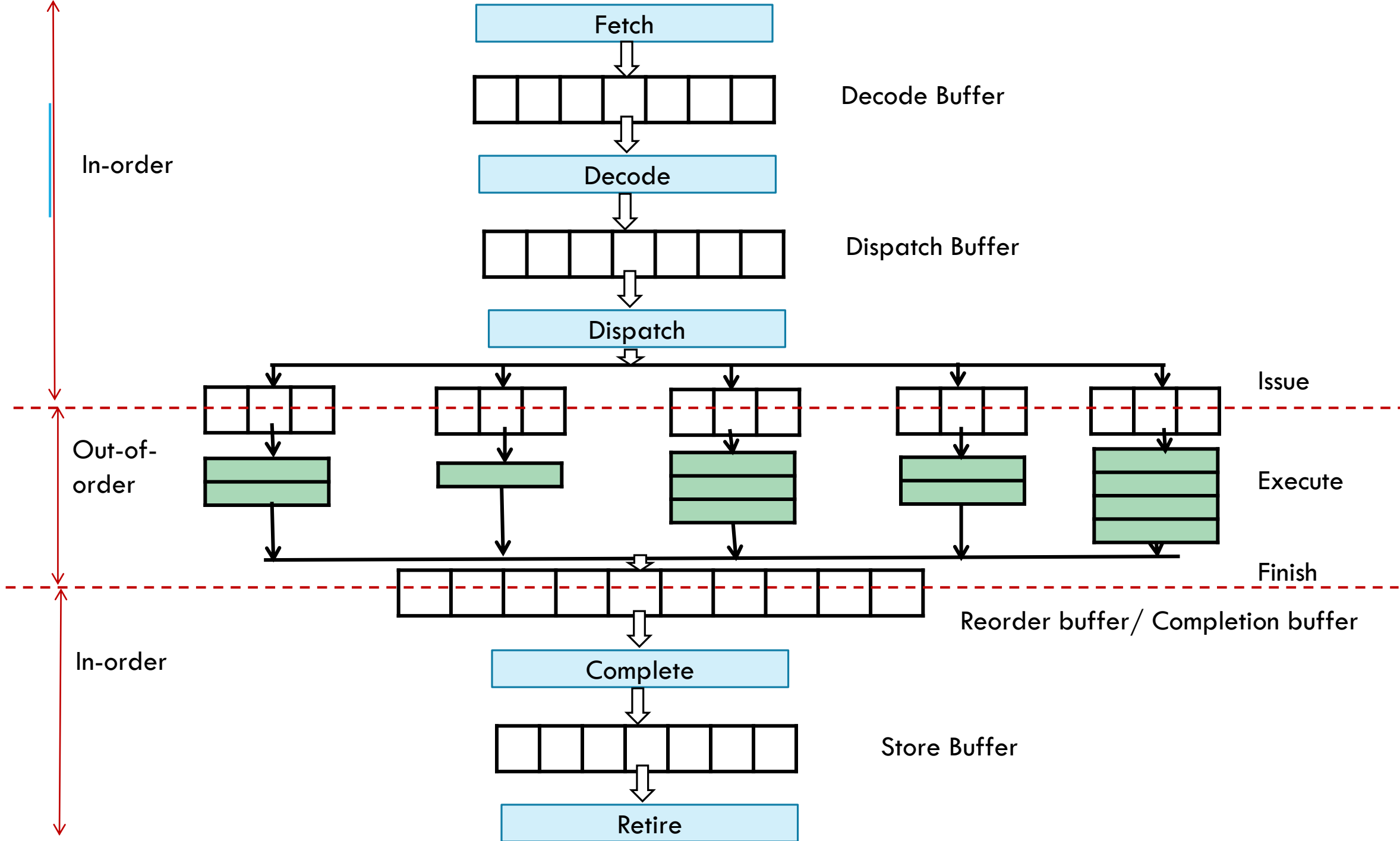
Pipeline Implementation:



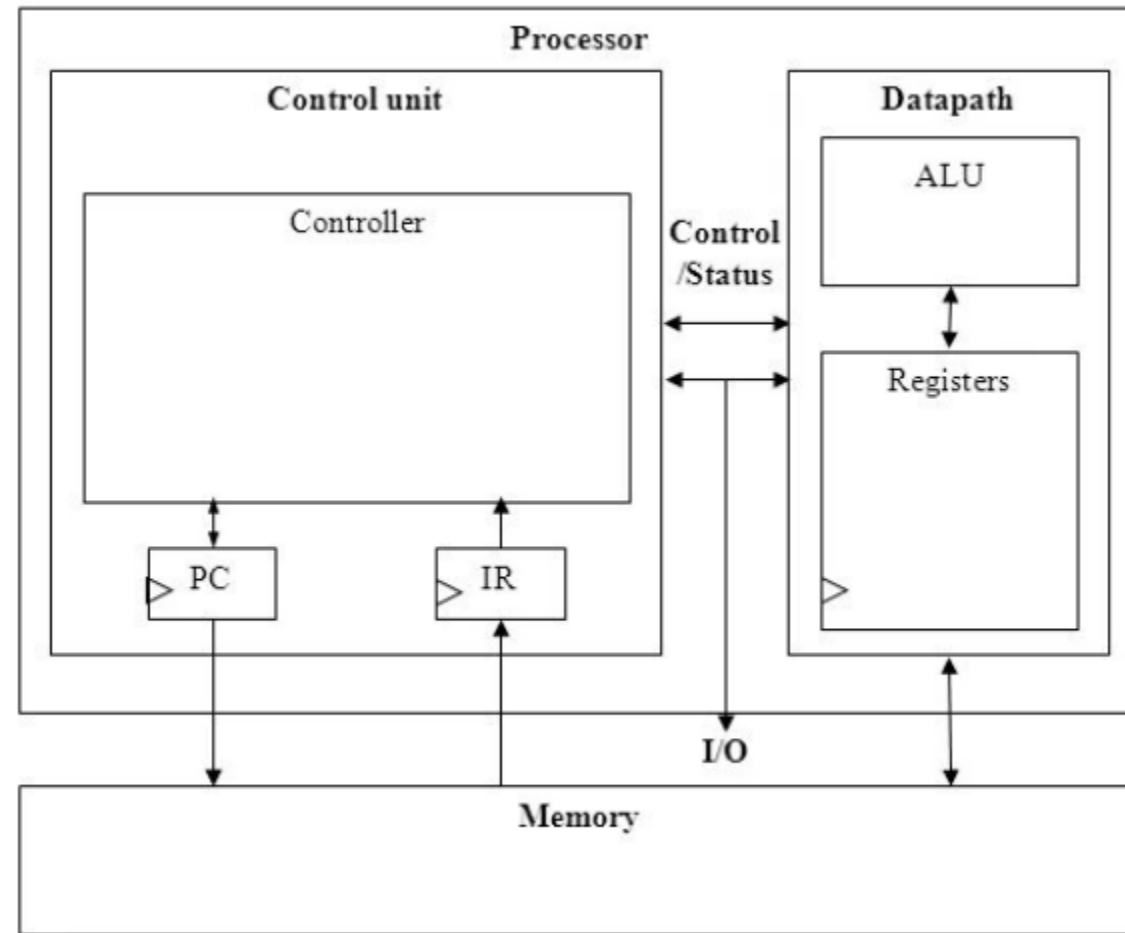
LDR R0, [R1]
LDR R0, [R1, R2]
STR R0, [R1, R2]
ADD R1, R1, R2

SCALAR PIPELINED ARCHITECTURES





CPU INTERNAL ARCHITECTURE



COMPUTING PARADIGMS

The Von Neumann Computer

Domain Specific Processors/ Application Specific Instruction Set
Processors (ASIP)

Application Specific Integrated Circuits (ASICs)

Reconfigurable Processors

COMPUTING PARADIGMS

The Von Neumann Computer – Temporal Computation
Reconfigurable Processors

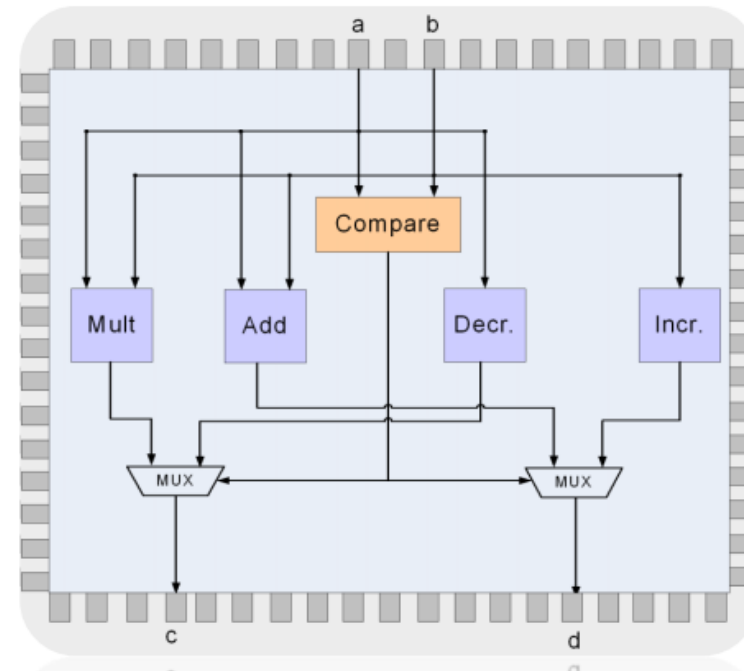
COMPARISON

□ Implementation of a VN computer

```
if (a < b) then
{
    d = a+b;
    c = a*b;
}
else
{
    d = a+1;
    c = b-1;
}
```

Ability to extract parallelism (or concurrency) from algorithm descriptions is the key to **acceleration using reconfigurable computing**

ASIC implementation:



RECONFIGURABLE COMPUTING UNIT

A **Reconfigurable Computing Unit (RCU)** is a specialized hardware component that allows **its internal architecture to be dynamically modified or reprogrammed** after manufacturing, enabling it to perform a wide range of computational tasks efficiently.

- FPGA (Field Programmable Gate Array),
- Coarse-Grained Reconfigurable Arrays (CGRAs),
- Adaptive Compute Acceleration Platform (ACAP)
- Runtime-Reconfigurable Processors, etc.